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Next item →

1. Consider the "state equation" describing the evolution of parameter values over time: $\theta_k = \theta_{k-1} + r_{k-1}$. In this equation, what is the role of r_{k-1} ?

1 / 1 point

- ☐ It is a random variable for which we compute a random sample value every filter iteration, which we add to the prior state estimate.
- ☐ It is a quantity that precisely describes how the parameter values change over time.
- ☒ It is a fictitious quantity that is never used directly by the xKF, but whose positive-definite covariance allows the xKF to adapt parameter values.
- ☐ It is a random quantity that enforces that model parameter values only degrade over time, and can never improve.

✔ Correct
Yes. This is correct.

2. Consider the "output equation" of the model describing parameters, $d_k = h(x_k, u_k, \theta, e_k)$. Which of the following are true? (Select all that apply.)

1 / 1 point

- ☐ This equation *must* model the same measurable output as the dynamic model of the system. That is, $d_k = z_k$.
- ☐ This equation describes d_k , which is a conceptual quantity only. It does not model any measurable quantity.
- ☒ This equation can model any physical measurement d_k that is a function of state, input, and parameters.

✔ Correct
Yes. This is true.

- ☒ This equation *may* model the same measurable output as the dynamic model of the system. That is, d_k may be equal to z_k .

✔ Correct
Yes. This is one possible use for this equation.

3. In addition to defining "state" and "output" equations for the parameters themselves, how must we modify the "state" and "output" equations for the model that describes the relationship between the dynamic system's input u_k and the dynamic system's output z_k ? (Select all that apply.)

1 / 1 point

- ☒ We must explicitly show dependence on the parameter vector θ in the state equation $f(\cdot)$ and the output equation $h(\cdot)$.

✔ Correct
Yes. This is true.

- ☐ We must make both output equations $d_k = h(\cdot)$ and $z_k = h(\cdot)$ describe the same measurement.
- ☐ We must use the same measurement-noise input to both output equations.
- ☐ We must use the same process-noise input to both state equations.